

FPGA design and implementation of Digital PWM technique for DC-DC Converters

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Abstract-- Digital Pulse Width Modulation (DPWM) is one of the most elementary parts in Digital Control Systems for the control of the power switch. Hence it has become an integral part of all System on Chips. It has also been broadly acknowledged as the control method in power converters. The principal focus of this paper is the design of FPGA based Digital controller for a DC-DC Buck convertor for SoC. The output voltage of a DC-DC convertor relies on the switching control circuit which is fundamentally an FPGA Digital controller-PWM.VHDL is employed for the coding of the Digital Controller, synthesis has been done using Xilinx XST 14.2, and Behavioral and Timing simulation has been handled by ModelSim. Also a power analysis using the XPower Analyzer is being carried out. After verification of the code, it will be downloaded onto a Spartan 3AN FPGA board.

Keywords: Systems on Chip, DC-DC Buck Converters, Digital Controller, Oscilloscope.

I. INTRODUCTION

The steady state response and the dynamic behavior of the DC-DC Converters have become the nucleus of research in Power Electronics. A number of control circuitry for these converters both analog and digital techniques have been proposed. But with the advent of highly precise Microprocessors and inbuilt functions of the FPGAs, digital control has gained large importance. Also efficiency, upgradeability and low sensitivity to component variations of the digital methods act as an added advantage.

The DC –DC Buck converters is a power converter that steps down the voltage from the supply to the load. The main parts of a digitally controlled Buck Converter are Buck Converter circuit, Analog to Digital converter, digital Compensation unit for the control of duty cycle and a PWM circuit as a voltage regulator.

Pulse width modulation (PWM) is a powerful method of controlling analog circuits with the aid of digital outputs [2]. PWM is employed in a wide range of applications, from measurement and communications to power control and conversions. The PWM bears significance in control systems like DC-DC converters where the PWM signals were used to change over unregulated DC voltage to regulated or variable DC voltage on the output side [5]. A PWM signal is not steady, the main parameter is its duty cycle D that is a part of PWM period and depicts the proportion of the on time to regular interval. The equation (1) describes the duty cycle as the following:

$$D = \text{on time} * \text{switching frequency} \dots (1)$$

where D ranges between 0 and 1 for Buck converter.

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The output voltage can be given by equation (2).

$$\text{Output voltage} = \text{Input voltage} * D \dots (2)$$

Hence the average output value depends on the duty cycle.

Various PWM architectures have been proposed in the late years which were able to produce fewer harmonics along with a high resolved output. The selection of the architecture depends on the application for which it is being used [5].

II. PROPOSED ARCHITECTURE

The main advantage of PWM lies in its low power and high resolution in switching circuits [3]. In conventional PWM architectures, a binary comparator compares the input value with the counter output and on its equivalency resets the S-R flip-flop thus generating variable duty cycles [5]. The main advantage of the proposed architecture would be low power and smaller circuit area owing to the replacement of the Binary counter and Comparator conventional blocks with the Gray counter and 1-hot encoder circuits respectively for the generation of the variable pulses for the control of DC-DC Buck Converters.

In the proposed architecture, a 4 bit Gray counter ticks on every positive edge of the clock. On encountering a “0000” at the counter output, the enable logic is made high which turns on the one-hot encoder thus acting as the reset pin for it. Thus the one-hot encoder circuit is on only when required thus reducing the switching power. The one-hot encoder circuit is similar to a ring counter that shifts the bits on receiving an enable signal. As shown in Figure 1, the MSB output of the one-hot encoder is given to the set pin of the RS flip-flop. Depending on the requirement of the duty cycle, the PWM control circuit resets the flip-flop thus producing a variable duty cycle.

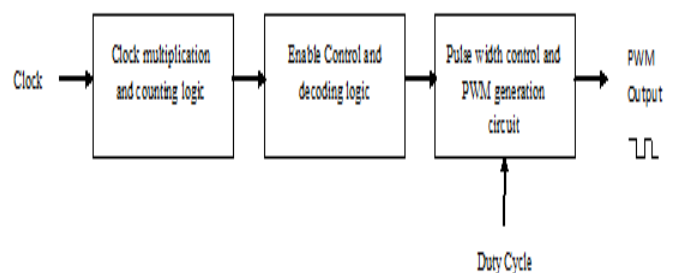


Fig 1: Proposed PWM architecture

Figure 2 shows the algorithm for the proposed architecture.

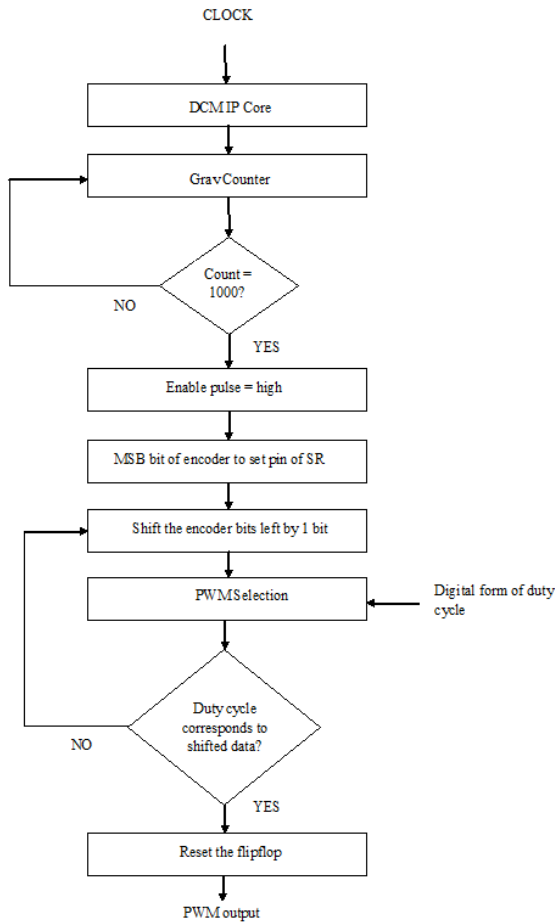


Fig 2: Algorithm for the proposed architecture

III. RESULTS

1. SYNTHESIS REPORT:

The synthesis report demonstrates the number of logic blocks and device utilization by the architecture. The following tables 1 and 2 shows the logic block usage and device utilization by the proposed architecture respectively.

Device name: xc3s50a-4tq144 (SPARTAN 3AN FPGA)

TABLE 1: MACRO STATISTICS OF LOGIC BLOCKS USED IN THE PROPOSED ARCHITECTURE

LOGIC BLOCKS	NUMBER OF BLOCKS USED
Registers	2
1 bit register	1
16 bit register	1
Multiplexors	1
1 bit 16 to 1 multiplexor	1

Device name: xc3s50a-4tq144 (SPARTAN 3AN FPGA)

TABLE 2: DEVICE UTILIZATION OF THE PROPOSED ARCHITECTURE

NAME	NUMBER USED	% USED
Slices	24	3
Slice filpflops	32	2
4 input LUTs	14	0
IOs	9	-
Bonded IOBs	9	8
GCLKs	3	12
DCMs	1	50

Comparison Area and Speed for different Spartan and Virtex FPGA Families

FPGA	Spartan 3A-DSP	Spartan-6	Virtex 5
Device	Xc3d1800a-4cs484	Xc6slx4-3tqg144	Xc5v1x20t-2ff323
Slices	24	21	32
Slice filpflops	32	10	25
4 input LUTs	14	7	18
IOs	9	9	9
Bonded IOBs	9	8	-
GCLKs	3	-	-
DCMs	1	-	-
Number of BUFG/BUFGCTRLs	-	3	3
Maximum Frequency	115.794MHz	171.183MHz	203.936MHz

2. RTL SCHEMATIC AND SIMULATION RESULTS:

The Figure 2 shows the RTL schematic of the proposed architecture after the synthesis of the design using Xilinx XST tool. Simulation of the architecture has been carried out launching ISim in the Xilinx behavioral implementation simulation.

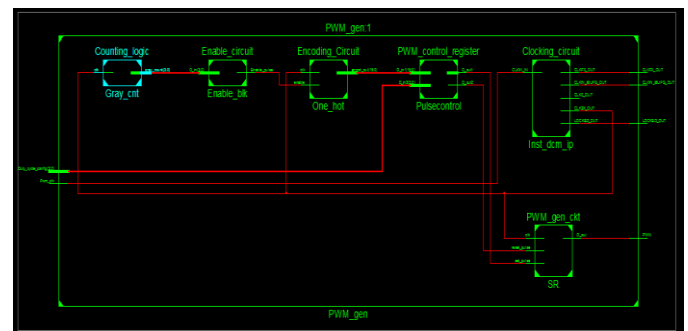


Fig 2: RTL Schematic of the proposed architecture

➤ With 25% duty cycle

The corresponding digital form of duty cycle = 0011
Output voltage obtained = $0.25 * 5 = 2.5V$.

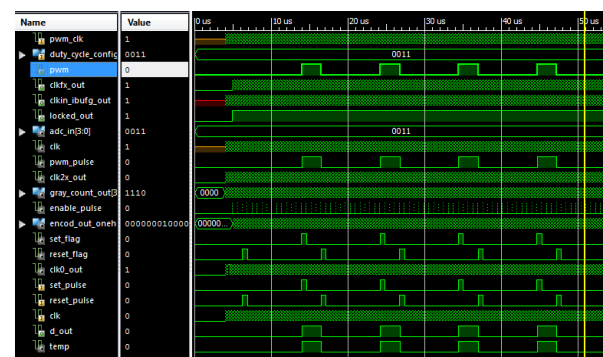


Fig 3: PWM output for 25% duty cycle

➤ With 75% duty cycle

The corresponding digital form of duty cycle = 1011
Output voltage obtained = $0.75 * 5 = 3.75V$

TABLE 3: DIGITAL INPUT ALONG WITH THE CORRESPONDING DUTY CYCLES AND OUTPUT VOLTAGES.

INPUT	DUTY CYCLE (%)	ON TIME(ns)	OUTPUT VOLTAGE
0000	6.25	78	0.3125
0001	12.5	158	0.625
0010	18.5	235	0.925
0011	25	310	1.25
0100	31.25	390	1.5625
0101	37.5	475	1.875
0110	43.75	546	2.1875
0111	50	630	2.5
1000	56.25	718	2.8125
1001	62.5	798	3.125
1010	68.75	876	3.4375
1011	75	947	3.75
1100	81.25	1026	4.0625
1101	87.5	1114	4.375
1110	93.75	1196	4.6875
1111	100	1273	5

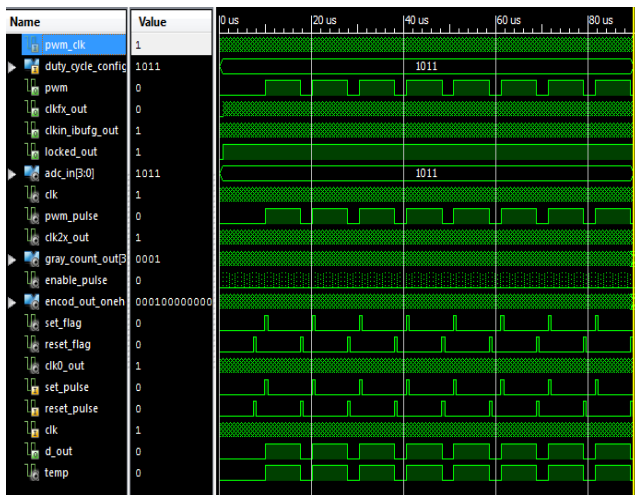


Fig 4: PWM output for 75% duty cycle

PWM output for various values of duty cycles is as shown in Figure 5.



Fig 5: PWM output for variable duty cycle

3. Result table:

Required switching frequency = 0.78MHz.

Number of counter bits = 4.

Hence clock frequency = $0.78\text{MHz} \times 2^4$
= 12.5MHz.

Min Duty Cycle = Min on time * switching frequency
= $80\text{ns} \times 1 / 1280\text{ns}$
= 6.25 %.

Min Output Voltage = Input Voltage * Min duty cycle
= 5×0.0625
= 0.3125V.

The following table 3 shows the values of the digital input along with its corresponding duty cycles, on times and output voltages. Assume the input voltage to be 5V.

IV. CONCLUSION

This paper proposes a novel Digital PWM architecture for DC-DC Converters. This architecture includes a DCM block for improving the switching frequencies. Also the Gray Counter and one hot encoder circuits act as the added advantage for low power consumption. Xilinx ISE 14.2 and ISim were used to design and simulate the architecture respectively. The design works good for frequencies upto 160MHz.

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